VIKISEWS

HOW FABRIC PROPERTIES AFFECT PATTERN HACKING RESULTS

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Different types of fabrics have different properties and characteristics depending on their intended use. While special-purpose fabrics prioritize high strength properties, everyday fabrics are more concerned with aspects like hygiene, crease resistance, shrinkage, and so on. Let's explore the essential properties of fabrics in more detail.

FABRIC WEIGHT

Fabric weight means the weight of one square meter of fabric. This measurement allows you to estimate the weight of your finished garment in advance. Fabric weight depends on the thickness of the warp and weft threads, as well as on how these threads are interlaced in the fabric.

Fabric weight can vary significantly, ranging from 12 to 900 g/m². It determines the fabric's intended use and the type of garments that can be sewn with it, such as blouse- and/or dress-weight, suiting, coating fabrics, and more.

Fabrics with a low fabric weight, such as chiffon (40–60 g/m²), are thin and semi-sheer. They are used for making dresses, blouses, and underwear, often as double-layered or lined garments.

Fabrics with a higher weight, such as blouse- and dress-weight materials of 125 g/m², are thicker than chiffon. However, they may still be see-through and thus not ideal for dresses. Therefore, these fabrics are mostly used for blouses. A dress made from such fabric will need a lining or an extra fabric layer. However, if you choose a cotton fabric with the same weight, it's quite possible that it will be opaque enough for a dress. Dress fabrics should not be see-through; otherwise, a lining is needed.

When choosing a coating fabric, it's important to understand that a higher fabric weight will result in a heavier finished garment. In addition to the weight of the main fabric, you also need to consider the weight of the interfacing, insulation, and lining, so this should be taken into account.

Fabrics with the same weight but different compositions can look different. For example, with the same weight of 305 g/m², denim and linen will look and feel quite different. Denim will be soft and lightweight, while linen will be heavyweight and stiff. The same fabric can look different in various colors. Light colors may appear transparent and see-through, while dark colors may be completely opaque.

Fabric weight doesn't reflect characteristics like wrinkling, pliability, elasticity, drapability, stretchiness, or durability. It also doesn't reflect how the fabric feels, whether it's soft or stiff, itchy or silky, or if it will shrink. Therefore, when choosing a fabric, you should not rely solely on the fabric weight. It's also important to ask the seller for additional information or even for videos of the fabric in question.

To make it easier to choose the right fabric for a specific project, you can refer to fabric weight charts that list different fiber compositions and their intended uses.

The values are averaged and are provided as recommendations.

FABRIC WEIGHT

Fabric composition	*Fabric type and use*	*Fabric weight, g/m²*
Cotton, cotton blends	*Cheesecloth*	*30–36*
	Batiste	*55–70*
	Percale	*65–122*
	Chintz	*80–100*
	Poplin	*90–150*
	Calico	*110–150*
	Gauze	*115–240*
	Lightweight sateen	*113–130*
	Heavyweight sateen (for robes, bed linen)	*130–150*
	Sateen jacquard	*135–140*
	Flannel	*about 160*
	Twill	*185–260*
	Cotton terry cloth	*200–500*
	Suiting	*320–360*
	Velvet, corduroy	*240–400*
	Lightweight denim	*305–390*
	Classic midweight denim	*400–440*
	Heavyweight denim	*470*

Fabric composition	Fabric type and use	Fabric weight, g/m²
Silk	Voile, crepe chiffon	25–40
	Silk gauze / chiffon	40–60
	Silk / silk satin	80–90
	Taffeta	110
	Blouse- and dress-weight fabrics	125
	Lining fabrics	125
	Crepe de chine	180–200
	Crepe back satin	230–240
	Satin	230–240
	Jacquard	110–310
	Raincoat fabrics	75–180
	Silk velvet	200–300
Wool, wool blends	Lightweight dress-weight fabrics	140–190
	Dress-weight and suiting fabrics	210–310
	Suiting	290–310
	Heavyweight suiting and lightweight coating fabrics	340–360
	Blended coating fabrics	450–600
	Coating fabrics	450–550
	Heavyweight coating fabrics	550–700
	Wool coating	450–700
	Double-faced coating	450–950
Linen, linen blends	Blouse- and dress-weight fabrics	120–165
	Lightweight suiting and dress-weight fabrics	150–180
	Heavyweight suiting fabrics	180–350
	Toweling linen	200–300
	Buckram	190–360
Viscose, viscose blends	Lining fabrics	70–160
	Viscose knits	160–260
	Lightweight viscose for blouses, shirts, dresses	about 200
Polyester	Fleece	130–340
	Dress-weight fabrics	140–250
	Suiting fabrics	200–350

Fabric composition	*Fabric type and use*	*Fabric weight, g/m²*
Knits: cotton, cotton blends	*Jersey knits*	*145–180*
	Quilted knit fabrics	*160–180*
	Interlock knits	*160–260*
	Ribbed knits	*170–350*
	Piqué knits	*180–260*
	Sweatshirting	*180–330*
	Terry cloth knit fabrics	*200–280*

FIBER CONTENT

The fabrics used in garment sewing can have various fiber content, including natural fibers (wool, silk, cotton, linen), artificial fibers (viscose, acetate, etc.), synthetic fibers (polyester, nylon, PAN, etc.), or blends.

Based on their fiber content, fabrics are divided into homogeneous, mixed, or heterogeneous:

Homogeneous fabrics consist of one type of fibers or thread. Those can be fabrics made entirely from cotton, pure linen, pure wool, etc. Fabrics are considered homogeneous if they contain up to 10% fibers of other types in addition to the main fiber.

Mixed fabrics have different fibers in their warp and weft threads that are blended during spinning. For instance, the fibers in the warp and weft can include wool blended with nylon, or linen fibers blended with polyester.

Heterogeneous fabrics have warp and weft made from different types of fibers. An example is a fabric with a cotton warp and linen weft, or a fabric made from natural silk twisted with triacetate thread.

The fiber content affects the hygienic properties of the fabric.

Hygienic properties show how the fabric interacts with the environment and include: breathability, air permeability, electrostatic properties, water resistance, vapor permeability, and thermal capacity.

Breathability refers to the fabric's ability to absorb moisture from the surrounding environment. Wool garments have the highest breathability. Linen fabrics, as well as natural silk, viscose, and cotton, also quickly absorb and release moisture. Synthetic fibers generally have lower breathability.

Air permeability determines the ventilating properties of the fabric. Low air permeability implies good resistance to wind. Air permeability largely depends on the fiber content, fabric weight, and fabric finish. Fabrics made from natural fibers with fine fibers often have higher air permeability compared to those made from solid synthetic fibers.

Electrostatic properties determine the fabric's ability to accumulate static electricity on its surface. Electric static charges are generated and dissipated continuously from friction. Synthetic fibers with low breathability tend to accumulate more static electricity.

Water resistance describes the fabric's ability to resist initial water penetration. This property is crucial for raincoat fabrics.

Vapor permeability largely depends on the breathability of the fibers, as well as fabric weight, weave type, and finish. In loosely woven fabrics, moisture vapors pass through the fabric pores, while in densely woven fabrics, vapor permeability is due to the high breathability of their fibers. Thus, sportswear is often made from synthetic fabrics with high air and vapor permeability due to their special weave and finish.

Thermal insulation properties indicate the fabric's ability to conduct heat, in other words, to change its temperature according to external conditions. Linen is considered the coolest fiber due to its high thermal conductivity, whereas wool is the warmest, owing to its low thermal conductivity. Wool's low thermal conductivity is due to the air-filled channels within its fibers.

SHRINKAGE AND WRINKLING

Shrinkage refers to the change in fabric dimensions due to moisture and heat treatment or washing. Measured as a percentage, it indicates how much a fabric may shrink as compared to its original size. All fabric types are subject to shrinkage, so it's vital to consider this when purchasing fabric.

Shrinkage occurs for several reasons:

- During the production process, the tension of the threads may not always be even, leading to elongation of the threads. When subjected to moisture and heat, these threads can then shrink.
- When the fibers in the thread absorb moisture, they swell; after drying, deformation can occur.
- Shrinkage can also be caused by the straightening of threads in one direction (either weft or warp) influenced by the other.

Natural fabrics like cotton and linen can shrink by 10–15%. Shrinkage in blended fabrics can vary up to 5%, depending on their cotton and linen content. Wool fabrics typically shrink by 8–10%. Synthetic fabrics generally do not shrink.

If the expected shrinkage of the fabric is minimal, it's advisable to add extra seam allowances when cutting. It's best to use fabrics with similar shrinkage rates for the outer garment layer and lining. Before cutting, it is recommended to pre-shrink the fabric.

Preshrinking is a type of heat and moisture treatment applied to a fabric through pressing and steaming. It is done to prevent distortion during sewing and to ensure that the garment retains its shape after washing or dry cleaning. Thus, make sure to preshrink the fabric prior to cutting.

Wrinkling refers to the fabric's tendency to form permanent creases and wrinkles when folded or under pressure. This affects the wear of the fabric, as the most significant wear and tear occurs along these folds and creases. The fabrics' ability to return to its original shape (to smooth out) after being folded or compressed is termed as crease resistance.

Wrinkling depends on the fabric weight. Heavyweight fabrics tend to have greater resilience, maintain their shape better, and wrinkle less. Fabrics with a loose structure, where threads can shift without much effort, exhibit significant wrinkling. The fiber content of the fabric also plays a role in its crease resistance. For instance, synthetic fibers in any fabric tend to reduce wrinkling. Thanks to elastic polyester fibers, the surface of the fabric rebounds quickly to its original shape, and creases smooth out spontaneously. Similarly, silk and wool are also considered to be crease-resistant. Cotton, linen, and viscose fabrics demonstrate the highest wrinkling potential.

DRAPABILITY AND STRETCHINESS

Drapability refers to the fabric's ability to take on a nice shape and form soft, flowing folds when suspended under its own weight. Drapability is influenced by the fabric's stiffness or softness, weight, fiber content, and the type of weave used. Generally, the stiffer the fabric, the better it holds its shape; the heavier the fabric, the more voluminous the folds will be. And the softer the fabric, the lighter and more delicate the folds are going to be.

Soft fabrics, such as chiffon, lightweight silk, satin, or batiste, drape more elegantly than stiffer materials. However, to maintain their shape, softer fabrics require a heavier weight.

On the other hand, stiffer fabrics (like taffeta, sateen, heavyweight silk, or denim) can be more challenging to drape due to their resistance to mechanical manipulation. However, if a stiff fabric is thin, such as organza, the draping process can be easier.

Therefore, soft and heavyweight fabrics or lightweight and stiff fabrics are ideal for draping.

The ***stretchiness*** of fabrics in different directions is essential to consider when you hack a pattern and decide on the type of cut, particularly for stretchy fabrics.

The fabric does not typically stretch along the selvedge (lengthwise grain), but it may be more stretchy on the crosswise grain, depending on the fiber content. Additionally, there is a bias direction, where the fabric stretches diagonally across the weave. When cutting on the bias at a 45-degree angle to the lengthwise grain, the fabric gains extra flexibility and stretch, beautifully following the shape of the body. You can cut different types of fabric on the bias grain, from silk to wool (dress-weight, suiting). The goal here is to achieve that fluid, graceful shape.

When hacking a pattern intended for bias-cutting, it is important to use some sewing tips. For example, you can let the garment pieces hang freely to true them after they stretch out and prevent any possible misalignment.

When working with stretchy (knit) fabrics, it's crucial to account for the stretch factor, which impacts the choice of the basic pattern and the ease allowances, especially when you deal with close-fitting garments without darts.

The ***stretch facto***r shows how stretchy the fabric is and significantly influences how well the finished garment will fit. Before sewing a project with a stretchy fabric like knits, it’s advisable to determine the fabric's stretch factor to find out how stretchy it is.

FRAYING AND THREAD SHIFTING

Fraying refers to the ability of the fabric threads to slip and unravel along the cut edges when subjected to mechanical stress. This characteristic varies depending on the fabric thread direction. The greatest fraying occurs along the warp, or weft, or at angles no greater than 15 degrees to the threads. Conversely, the fabric cut on the bias (at a 45-degree angle) experiences minimal fraying.

The degree of fraying in fabrics largely depends on their fiber content. The weave type also plays a significant role in fraying. Thus, fabrics with a satin weave experience three times as much fraying compared to a plain weave. Fabrics with satin weaves featuring larger thread overlaps show the highest tendency to fray, while plain weaves exhibit the least. A lower thread count in one of the directions can increase the fraying in the crosswise direction.

The extent of fraying influences the choice of cut. To protect the edges from fraying, additional finishing techniques such as overlocking, reinforcement with fusible interfacing, bias binding, and pinking cuts may be applied. Special sewing techniques, such as flat-felled seams, French seams, or a double-fold hem, can also be used to enclose raw edges within the seam.

Shifting of threads occurs when the warp and weft threads of the fabric move against each other.

This can typically be seen in the seam areas of finished garments (like the dart seams, center back seams, sleeve-to-armscye seams, and side seams). The bad choice of fabric or garment shape can lead to excessive thread shifting, compromising the garment's appearance and durability. Fabrics prone to shifting (loosely woven and with a low weight) are not advisable to be used in close-fitting garments, as it may cause early wear. To reduce thread shifting in the seam areas, you can use smaller stitches. This can enhance friction between the fabric threads, thereby minimizing their ability to move.

PATTERN AND COLOR

All fabrics can be divided into several main types based on their patterns:

Stripes come in various width, complexities, and frequencies. They can be facing in different directions and can be single or multi-colored. Often, the striped patterns are aligned with the lengthwise grain on a white or colored background.

Polka Dots are represented by black, white, or colored circles of various sizes scattered across a white or colored background. This pattern is typically used for women's garments, particularly in dresses and blouses.

Checks exhibit a wide range of patterns. They can be simple, formed by the intersection of uniform stripes, or complex, made with stripes of different width and colors.

Floral patterns can feature flowers or bouquets distributed across a white or colored background.

Panel prints are large motifs, often featuring floral designs, situated in the center of the fabric against a variety of colored backgrounds. Panel fabrics should be cut in a special way to ensure that the pattern remains intact wherever possible.

Small print and large print patterns consist of geometric shapes of different forms and sizes, featuring either tiny or large motifs and various thematic designs scattered across the fabric.

Each pattern has a repeat, or rapport, which defines how the design is repeated. Therefore, choosing a patterned fabric involves some specific considerations. For fabrics with large designs, it's advisable to opt for simple garment designs with fewer seams. Attention should be paid not only to the size of the pattern but also to its direction.

It's essential to plan the cutting of garment pieces so that the pattern repeat aligns correctly in the finished garment. At the dart seams, the pattern should maintain its longitudinal position and join seamlessly in a "herringbone" manner with the other side.

Patterns that are miniature, small, or medium-sized and densely fill the background are considered easier to work with. Large patterns located in prominent areas (like the bust or hips) can visually distort the figure.

In the case of printed fabrics, where the right side typically displays bright and clear designs while the fabric surface is smoother and slightly shiny, certain design elements like princess seams or godet pieces may not stand out as effectively as you wish. Draping looks best on solid-colored fabrics or those dyed uniformly on both sides. Draping on fabrics with very small printed patterns tends to be nearly invisible.

The ***texture*** of a fabric refers to its overall appearance and feel. It encompasses whether the fabric appears smooth or rough, textured or flat, soft or stiff, shiny or matte, sheer or opaque, and more. The fabric texture often influences the cut of a garment, as well as our perception of its shape and even its color.

Texture can enhance the color effect of a fabric. For instance, the combination of soft and vibrant shades will stand out even more when using fabrics with contrasting textures.

Shiny fabrics reflect light, making colors appear more vibrant, while matte fabrics absorb light, giving them a darker appearance. Soft and flowing fabrics are generally more loose, with greater spacing between their threads or fibers compared to heavier fabrics. Heavyweight fabrics retain their shape excellently and are well-suited for simple garment shapes with sharp lines. Textured fabrics are usually thicker than smooth ones, adding volume and softening harsh lines. Smooth fabrics can vary significantly, ranging from lightweight and sheer to dense and heavyweight.

Since we perceive textures not only visually but also through touch, they can be divided into raised, grainy, smooth, and slippery textures. Fabrics with pronounced textures add volume, while a smooth surface creates an impression of lightness and visually reduces bulk. The shape of a garment is directly influenced by the texture of the fabric used. It is impossible to achieve crisp, sharp lines with soft, airy fabrics, and vice versa. For example, soft and pliable fabrics are more appropriate for rounded and oval shapes, while firmer and denser fabrics like wool coating or leather are better suited for straight silhouettes. This means you should consider both the shape of the garment and the intended texture.

Using different textures within a single design allows you to highlight the unique features of each surface.